GTFShift for bus lane prioritization using GTFS and OSM data

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Abstract

Identifying the road network segments with the highest bus frequency per hour can help in planning and prioritizing the implementation or expansion of bus lanes. GTFShift R package uses the General Transit Feed Specification (GTFS) to perform this analysis, allowing for an aggregated overview of the transit supply over a given territory.

Keywords: Bus lanes, GTFS, Transit planning, Network prioritization, Decision support tool, R package.

1 Questions

Bus lanes (or transit priority lanes) have the potential to improve the reliability of bus operations by limiting the negative impacts of traffic congestion (Cesme et al., 2018), reducing the variability of travel times (Surprenant-Legault & El-Geneidy, 2011), and increasing the average commercial speed – which can ultimately be used to increase service frequency (Basso et al., 2011). These benefits are amplified when there is a continuous bus lane network, with the greatest travel time savings at intersections (Truong et al., 2015), particularly if vehicles obtain right-of-way at traffic lights. However, introducing them on road infrastructure affects other modes, such as private vehicles, which may experience a decrease in the level of service due to reduced allocated space (Arasan & Vedagiri, 2010), potentially jeopardizing public acceptance (Batty et al., 2014).

Common criteria for implementing bus lanes include:

- Frequency of buses (and trams) per hour and direction, at a peak hour;
- Number of lanes in the same direction;
- Existing traffic conditions;
- Existing bus lanes in the area (from a network continuity perspective).

For instance, NACTO reccomends that dedicated bus lanes should be applied «on major routes with frequent headways (10 minutes at peak) or where traffic congestion may significantly affect reliability» (NACTO, 2016). Nevertheless, the decision should be based on a more detailed analysis of the network, including the existing traffic conditions, the number of bus stops, and the demand for public transport in the area.

Vuchic (1981) suggests that bus lanes should be implemented based on the number of lanes, vehicle volume per lane, and bus frequency per hour. For instance, the minimum bus frequency for a street with 2 lanes in the same direction and 400 veh/h/lane is considered to be 12 bus/h to justify its implementation, according to the author.

This paper focuses on estimating hourly bus frequency for each road network segment in urban areas, using General Transit Feed Specification (GTFS) data, an international transit information standard, which is widely available from many transit agencies. Identifying high-frequency corridors is essential for prioritizing and supporting new bus lane implementation.

For this, we present `GTFShift`, an open-source R package (Matos & Félix, 2025) that extends the functionality of other R libraries that deal with GTFS, such as `tidytransit` (Poletti et al., 2025) and `gtfstools` (Herszenhut et al., 2025). This tool helps prioritize bus lanes in a given territory to improve transit efficiency.

2 Methods

GTFShift is an open-source R package that provides methods for the entire workflow of bus network frequency analysis. It can be used to get GTFS data for a given territory, and estimate the aggregated bus frequency per road network segment at each hour of a given day.

For this analysis, two main methods are used:

- unify(): Merges multiple GTFS feeds into a single entity, enabling an aggregated analysis. This method is relevant, for instance, when considering more than one bus operator in the same territory. It extends the functionality of gtfstools::merge_gtfs() by enabling the identification of transfers between agencies.

- get_route_frequency_hourly(): Calculates the hourly frequency of bus routes for a given day, considering its geometries. It assumes the time of departure at the first stop as a reference for each trip geometry. By default, it estimates the next business Wednesday, relevant for the peak hour. The overline parameter enables the aggregation of bus routes that share common line segments, returning a sum of frequencies per road segment, using stplanr::overline2() (Lovelace & Ellison, 2018). Optionally, using use_osm_routes parameter, it retrieves the geometries from *OpenStreetMap* (OSM) by matching the tag gtfs:shape_id, overwriting the original GTFS shapes.txt.

The parameter use_osm_routes is particularly useful if the GTFS shapes do not share the same geometry. For instance, if the edges of the lines do not overlap or do not follow the same route-over-the-road – which is very common, even besides GTFS recommendation – geometries might not be aggregated correctly, causing inconsistent results. By relying on a common road network, such as OSM, it is possible to overcome this issue and aggregate the bus routes correctly. Nevertheless, the gtfs:shape_id OSM tag is only present in 3.5% of the mapped bus networks. Alternatively, the osm_shapes_match_routes() method can be used to match GTFS shapes with OSM networks, considering their geometrical resemblance.

The case study is Lisbon, Portugal, where the bus network is operated by *Carris* and *Carris Metropolitana*, respectively, the urban and sub-urban bus operators in the area.

3 Findings

We load the GTFShift R library, the static GTFS feeds for both bus agencies, and unify them.

```r
library(GTFShift)

# loaf GTFS feeds
gtfs_carris = load_feed("https://gateway.carris.pt/gateway/gtfs/api/v2.8/GTFS") # 114 MB
gtfs_carrismetro = load_feed("https://api.carrismetropolitana.pt/gtfs") # 638 MB
```

```r
# unify the feeds in a single GTFS feed
gtfs_lisbon = unify(gtfs_carris, gtfs_carrismetro)
```

Next, the hourly bus frequency is estimated for each route, for the next business Wednesday, and the geometries are overlined, considering the correspondent OSM routes.

```r
# OSM query for bus networks in Lisbon
library(osmdata)
query = opq("Lisbon")  |>
        add_osm_feature(key = "route",
                        value = "bus") |>
        add_osm_feature(key = "network",
                        value = c("Carris", "Carris Metropolitana")) # operators
```

```r
# get hourly frequency by shape
frequencies_segment = get_route_frequency_hourly(gtfs = gtfs_lisbon,
                                                 overline = TRUE,
                                                 use_osm_routes = query)
```

The results, presented in Figure 1, allow for the identification of the road segments with higher bus transit frequency, at morning peak hour.

```r
# filter for the peak hour 8-9am
frequencies_segment_8 = frequencies_segment |>
  dplyr::filter(hour == 8) # the hour with the highest frequency

mapview::mapview(
  frequencies_segment_8,
  zcol = "frequency",
  lwd = "frequency",
  layer.name = "Bus frequency"
)
```

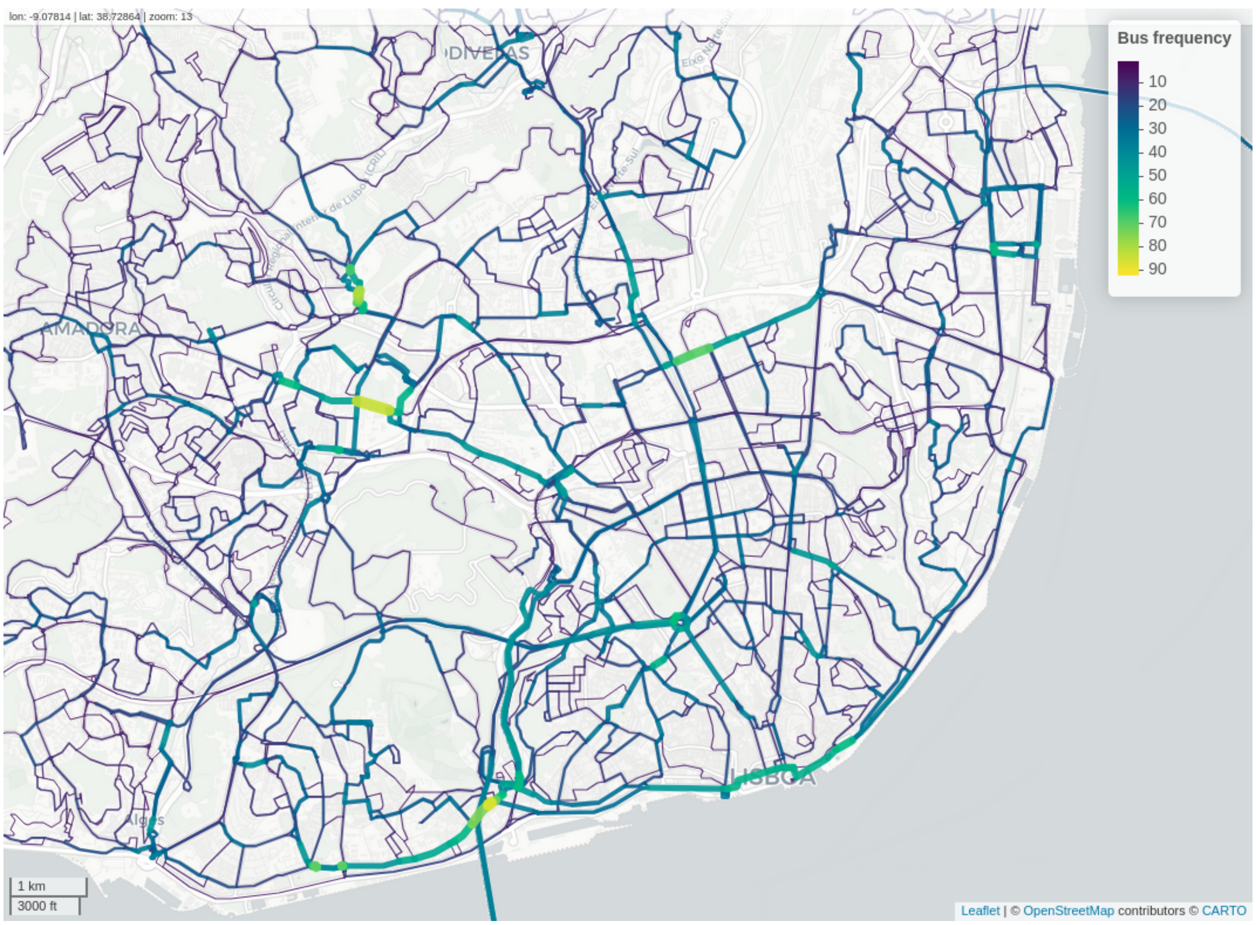

Fig. 1. Road segments with higher bus frequency, in Lisbon, at 8 am. The larger and yellowish lines represent the road segments with higher hourly bus frequency. Carto basemap.

By relying on OSM network data, this method allows to cross with other information available in this open geographical database. For instance, we can easily compare the results with existing bus lanes and identify high-frequency segments lacking dedicated infrastructure, which should be prioritized. Note that for two-way segments, frequency is aggregated.

Additionally, filtering the road network segments by the number of lanes in the same direction (two or more), can further support decisions on implementing bus corridors. However, policies may also favour bus lanes on single-lane streets, for example, to create transit-priority streets.

Further enhancements of the method may include passenger capacity per hour instead of just frequency. Routes with articulated or double-deck buses (around 100 passengers) could be given more weight than neighbourhood minibuses (up to 25 passengers).

This method supports not only static bus lanes but also dynamic bus lanes, by analyzing hourly frequency – providing evidence for decision on which hours to allocate road lanes for bus.

The presented replicable approach offers a feasible methodology to generate evidence for decision-making on the implementation or expansion of bus lane networks. Required data is commonly available for most cities, as GTFS feeds are widely used by public transport agencies.

We acknowledge that this evidence alone may not be sufficient to justify bus lane implementation, as it does not account for other factors such traffic conditions, roadway capacity, physical limitations, or public acceptance.

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4 Reproducibility

For this article, we used `GTFShift` version v0.7 and `R` version v4.5. To reproduce the results presented in this paper, refer to U-Shift/busclar code repository.

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